LAUNCHPATH AI

Product Requirements Document

Version 1.0 | Approved scope baseline

Your Intelligent Placement Readiness Companion

CAPSTONE PROJECT • INDIA • ZERO-BUDGET MVP

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1. Executive summary

LaunchPath AI — LaunchPath AI is a zero-budget-first, AI-assisted placement readiness companion for final-year students and fresher job seekers in India. It brings profile and resume review, role discovery, adaptive diagnostics, explainable readiness scoring, weekly learning plans, practice, interview simulation, job tracking and progress reporting into one responsive workflow.

Problem — Students typically prepare across disconnected tools: resumes in one place, aptitude practice elsewhere, scattered DSA resources, informal mock interviews and spreadsheets for applications. They lack a clear view of which target role fits, what skill gaps matter most, and what to do next. Generic roadmaps do not adapt to a learner’s baseline or weekly progress.

Product response — LaunchPath AI turns a student's profile and chosen target roles into an evidence-informed preparation loop: understand → assess → prioritize → practice → review → reassess. AI is assistive, not authoritative; transparent rules, curated resources and manual paths keep core journeys usable without paid APIs.

MVP principle — Deliver a dependable, demonstrable end-to-end student journey before adding breadth. Scores describe observed preparation evidence and are not hiring probabilities or guarantees.

2. Product vision, goals and non-goals

Vision — Make placement preparation more structured, explainable and actionable for every student, regardless of access to paid coaching or AI services.

Goals — 1) Help users select one primary role and up to two backups. 2) Identify strengths and gaps using resume evidence plus a role-based diagnostic. 3) Produce a transparent readiness score out of 100 with category breakdowns. 4) Convert gaps into stable weekly milestones and adaptive tasks. 5) Support repeatable aptitude, coding and interview practice. 6) Track applications and show progress through a downloadable report.

Non-goals — No hiring-probability prediction; no custom foundation-model training; no full online coding judge; no employer marketplace or college administration portal; no paid API dependency; no SMS/WhatsApp messaging; no guaranteed live job feed; no claim that AI output is verified unless supported by a cited source.

3. Target users and personas

Primary persona — Final-year student — A student with 3–12 months before placement season, a partially complete resume and uncertain role direction. Needs a clear baseline, practical weekly plan and beginner-friendly explanations.

Secondary persona — Fresher job seeker — A recent graduate applying to entry-level roles. Needs focused revision, interview practice, application organization and feedback that converts into next actions.

Assumptions — Initial launch is English-first and India-focused. Users may have inconsistent connectivity, low-end devices and limited budgets. The MVP supports English UI; Hindi and additional languages are architectural extensions rather than launch blockers.

4. User journeys and workflows

First-use journey — Landing/guest preview → create account or continue in fictional demo → consent choice → onboarding → upload PDF or enter profile manually → select primary role and up to two backups (or review top-three recommendations) → complete adaptive diagnostic → view readiness score and evidence → accept weekly plan.

Weekly learning loop — Open dashboard → review weekly milestone → complete curated lesson/tutor topic → practice aptitude or coding → complete interview session → receive rubric feedback → mark tasks complete → score and plan update with visible reason → weekly reassessment.

Application journey — Review curated/demo job or add a role manually → save → update status (Saved, Applied, Assessment, Interview, Offer, Rejected, Withdrawn) → add dates/notes → calendar view and reminders → include activity in report.

Fallback journeys — PDF parsing failure → manual profile. AI unavailable or consent declined → deterministic scoring, curated resources and template feedback. Voice unsupported → text interview. Job feed unavailable → curated demo jobs/manual entries. Cloud unavailable → local demo mode with clear persistence warning.

5. Functional requirements

FR-01 Accounts & guest — P0 — Email/password authentication; Google login only if a free provider is available. Guest mode uses fictional, clearly labelled sample data and cannot expose another user's data.

FR-02 Consent & privacy — P0 — Before each external AI analysis, show purpose, data categories, provider/destination and explicit opt-in. Decline path uses local rules/manual workflow. Allow revoke/delete/export where supported.

FR-03 Profile & resume — P0 — Profile editor for education, skills, projects, experience, preferences and links. PDF upload with extraction preview and user confirmation; parsing failure never blocks manual entry.

FR-04 Role discovery — P0 — Manual role selection plus optional recommendations. Show top three ranked roles with fit evidence, gaps, assumptions and confidence/coverage caveats. One primary + up to two backups.

FR-05 Diagnostic — P0 — Role-based adaptive quiz covering role-relevant concepts. Start with baseline items; select follow-up difficulty/topics based on responses. Save answers and completion state.

FR-06 Readiness engine — P0 — Produce score /100 and category scores with factor-level explanations, data freshness and next actions. Clearly label as preparation indicator, not hiring probability.

FR-07 Roadmap — P0 — Generate weekly milestones and tasks from gaps, available hours and role. Keep milestone structure stable; adapt task ordering/difficulty after progress or reassessment.

FR-08 Learning & tutor — P0 — Curated links/resources with level, topic and estimated time. Beginner-first tutor with topic-based learning memory only after consent; provide rule/curated fallback.

FR-09 Aptitude — P1 — Quantitative aptitude and logical reasoning practice; explanations, attempts and topic-level accuracy.

FR-10 Coding practice — P1 — Programming fundamentals and DSA questions, progressive hints, reference solution, complexity analysis. Lightweight editor optional; no full judge. User self-reports execution or uses sample test cases.

FR-11 Interviews — P1 — HR and technical mock interviews with Quick 5, Standard 10, Deep 15–20 and Custom presets. Curated questions plus AI follow-ups. Text-first; browser voice where supported and explicitly permitted.

FR-12 Evaluation — P1 — Rubric-based feedback: relevance, structure, correctness, clarity and examples. Explain score, quote short evidence, give one improvement task. Avoid unsupported claims about emotion/personality.

FR-13 Jobs & tracker — P1 — Matching against profile and role; live listings only from reliable permitted free sources. Otherwise curated demo listings and manual entries. Track all seven agreed statuses.

FR-14 Calendar & reminders — P1 — Calendar dates, manual due dates and in-app reminders only. No SMS/WhatsApp. Respect timezone and notification permissions.

FR-15 Analytics & reports — P1 — Dashboard trends for readiness, category scores, practice, roadmap completion and applications. Downloadable professional PDF with date range and methodology.

FR-16 Localization — P2 — English default; structure strings for Hindi and extensible language support.

FR-17 Deployment & offline fallback — P0 — Free public deployment where free-tier terms allow; local demo fallback. Communicate service state and persistence limitations.

6. Feature specifications

Resume analysis — Extract text from PDF client-side or through a permitted free parser. Show extracted fields before saving. Flag missing sections and evidence gaps; never invent experience or rewrite claims as facts.

Role fit — Use transparent role-skill mapping and weighted profile evidence. Return top three only when minimum profile completeness is met; otherwise ask for missing profile fields. Report matching evidence, missing evidence and suggested validation tasks.

Adaptive assessment — Question metadata includes role, topic, difficulty, answer key, explanation and source/license. Use a small deterministic branching rule (e.g., correct answer increases difficulty; incorrect answer selects prerequisite item).

Readiness score — Hybrid score from verified profile evidence, diagnostic performance, practice consistency and interview rubric. Each category displays numerator/denominator or contributing signals, data timestamp and confidence/coverage.

Roadmap — Use templates by role and skill gap. Keep weekly outcomes stable; replace or reorder tasks, not the overarching milestone, unless user changes target role or availability.

Tutor — Explain from first principles, define jargon, give one worked example, then a small practice check. Retain topic-level progress and misconceptions, not unnecessary raw conversation history.

Jobs — Do not scrape restricted sites or promise completeness. Show source and last checked date for live data; label demo/curated entries. Manual job creation always available.

PDF report — Include profile snapshot, target roles, readiness and category scores, score methodology, completed tasks, practice/interview highlights, applications and recommended next week. Exclude sensitive data not needed for preparation.

7. Readiness score methodology

Purpose — A descriptive preparation index to prioritize learning. It is not a probability of getting hired, an employer assessment or a guarantee of placement.

Suggested weights — Resume/profile evidence 20%; role diagnostic 30%; aptitude & reasoning 15%; coding/technical practice 20%; interview performance 10%; consistency & roadmap completion 5%. Total 100%.

Calculation — For each category, normalize available evidence to 0–100 using documented rules. Weighted total = Σ(category score × category weight). If a category has insufficient evidence, mark it 'Not assessed' and renormalize only across assessed categories; prominently show coverage (e.g., 4/6 categories assessed). Never silently assign zero for missing data.

Category examples — Profile: required sections, project evidence, role-skill coverage. Diagnostic: accuracy adjusted for difficulty and topic coverage. Aptitude: recent topic accuracy with minimum attempt threshold. Coding: completion, correctness from self-check/sample cases, explanation and complexity understanding. Interview: rubric average across completed answers. Consistency: completed planned tasks divided by due tasks, capped to avoid gaming.

Safeguards — Show factor explanations and data recency. Avoid false precision (display integer plus coverage). Reassess after meaningful new evidence or weekly check-in. Store score snapshots to show trend; allow correction of profile data. Prevent a score from being used as an automated rejection decision.

Category	Weight	Evidence examples
Profile / resume	20%	Completeness, projects, role-skill evidence
Diagnostic	30%	Accuracy, difficulty, topic coverage
Aptitude	15%	Recent quantitative + reasoning accuracy
Coding / technical	20%	Practice, sample checks, explanation
Interview	10%	Rubric average across answers
Consistency	5%	Due tasks completed; capped

8. AI architecture and fallback strategy

Architecture — Responsive web client → application/API layer → auth and data store → deterministic readiness/roadmap services → optional AI adapter → curated content catalog → reporting/export. Keep AI provider behind an adapter interface and isolate prompts from business rules.

AI uses — Optional resume-language suggestions, role-fit narrative, tutor explanations, interview follow-ups and rubric commentary. Inputs should be minimized and structured; outputs validated against schemas and treated as suggestions.

Lightweight ML/NLP — Use PDF text extraction, keyword/skill normalization, TF-IDF or simple similarity where feasible, and deterministic role-skill mappings. No custom model training. Explain ranking features.

Fallback matrix — AI unavailable/consent declined → rule-based matching, templates and curated resources. PDF parsing fails → manual profile. Voice unsupported → text. Job feed unavailable → curated demo/manual entries. Cloud unavailable → local demo mode.

Zero-budget assumption — Use open-source libraries and free tiers only after checking current limits, commercial/educational terms, data retention and regional availability. Keep core paths functional without external AI or live job APIs.

Reference flow: Responsive web client → application/API layer → auth + user-scoped data store → deterministic role-fit/readiness/roadmap services ↔ optional consent-gated AI adapter → curated content, practice bank and report export.

9. Data model and privacy

Core entities — User; ConsentRecord; Profile; ResumeDocument metadata; Skill; Role; RoleSkill; Assessment; Question; Attempt; ReadinessSnapshot; Roadmap; Milestone; Task; Resource; TutorTopicProgress; PracticeAttempt; InterviewSession; InterviewAnswer; RubricEvaluation; Job; Application; Reminder; Report.

Data principles — Collect minimum necessary data; separate account identifiers from analytics where practical; encrypt in transit and use provider encryption at rest; apply access controls and ownership checks to every record; avoid storing raw resumes after extraction unless needed and consented.

Consent record — Store consent purpose, provider, data categories, timestamp, policy version and revocation. Consent must be specific per external analysis, not a blanket hidden opt-in.

Retention & deletion — Define retention before launch; provide delete/export path within platform capabilities. Remove uploaded files after parsing when feasible. Do not send resume, contact details or interview audio externally by default.

Security — Validate file type/size, sanitize rendered content, protect secrets in environment variables, use HTTPS, rate-limit exposed endpoints where supported, prevent IDOR with user-scoped queries and never place API secrets in client code.

10. UI/UX principles

Principles — Professional, corporate, calm, responsive and accessible. Use clear hierarchy, plain-language labels, visible progress and explainable states. Avoid gamified pressure or shame.

Information architecture — Dashboard; Profile & Resume; Roles; Assessment; Roadmap; Learn/Tutor; Practice; Interviews; Jobs/Tracker; Calendar; Reports; Settings/Consent.

Interaction rules — One clear primary action per screen; autosave with status; confirm destructive actions; show empty/loading/error states; use mobile-first layouts; provide keyboard focus and sufficient contrast; distinguish sample data from user data.

Visual direction — Deep navy/ink base, clean white surfaces, restrained teal/blue accent, warm amber for attention, consistent typography, spacing and icon style. Avoid overloading dashboards with charts.

11. Non-functional requirements

Performance — Core dashboard should load quickly on a mid-range mobile device and modest network; paginate or lazy-load histories. Set an MVP target of <3 seconds for primary shell under normal test conditions.

Reliability — No data loss on refresh for persisted actions; clear local-demo warning if persistence is local-only. Graceful degradation for every optional integration.

Accessibility — Target WCAG 2.1 AA practices: semantic headings, labels, keyboard navigation, focus visibility, contrast and non-color status cues.

Compatibility — Current major Chromium, Firefox and Safari browsers; responsive from 360px mobile width upward.

Maintainability — Modular feature boundaries, typed schemas where stack supports it, reusable UI components, environment-based configuration and setup documentation.

Privacy & security — Explicit consent, least privilege, secure secrets, input validation, ownership checks and transparent retention.

12. Acceptance criteria

End-to-end — A new user can create/enter a demo account, complete profile manually, choose a role, take a diagnostic, see a readiness score with explanation, receive a weekly plan, complete a task, practice one aptitude/coding item, complete a text mock interview, add a job application, and export a report.

Fallbacks — With AI disabled and no external job feed, all core flows remain usable. PDF failure offers manual entry. Voice unsupported shows text mode. Cloud outage presents local demo mode without implying cloud persistence.

Scoring — Category weights sum to 100%; missing evidence is marked not assessed; coverage is visible; score wording explicitly disclaims hiring probability.

Privacy — External AI calls are blocked until explicit purpose-specific consent. User can revoke consent. User A cannot read or modify User B's records.

Quality — No broken primary navigation, no dead primary buttons, mobile layout works at 360px, errors are actionable, and build/deploy instructions are documented.

13. Risks, dependencies and mitigations

Scope overload — Risk: ten-day schedule exceeds available 20–30 hours. Mitigation: vertical slice first; P0 core flows before P1; use curated data/templates; defer optional voice, Google auth and live jobs.

Free-tier changes — Risk: provider quotas or terms change. Mitigation: adapter architecture, local mode, no paid dependency, verify terms before deployment.

AI hallucinations — Risk: invented resume claims, unreliable job facts or generic feedback. Mitigation: cite inputs, schema validation, user confirmation, source timestamps and deterministic fallbacks.

PDF parsing variability — Risk: scanned or complex layouts fail. Mitigation: extraction preview, size/type limits and manual entry.

Data privacy — Risk: sensitive resume/interview data exposure. Mitigation: explicit consent, minimization, deletion, user-scoped access and no client-side secrets.

Schedule — Risk: feature breadth dilutes quality. Mitigation: prioritize integrated flow, reserve Day 9 QA and Day 10 hardening; freeze features after Day 8.

14. Success metrics and v1.0 limitations

MVP success — A first-time tester completes the core journey without assistance; ≥90% of scripted critical-path tests pass; no critical privacy/security defects; score methodology is understandable to ≥4/5 test users; app works in AI-off demo mode.

Product metrics — Onboarding completion; profile completeness; diagnostic completion; roadmap task completion; weekly reassessment rate; practice attempts; interview completion; application status updates; report downloads; fallback activation/error rate.

Interpretation — Metrics are product usability and engagement indicators, not placement outcomes. Do not claim placement uplift without a properly designed study.

v1.0 limitations — Small curated question/resource bank; no full coding judge; no guaranteed live jobs; limited company-specific data; voice depends on browser support; English-first; lightweight local NLP; no custom model training; local mode may not sync.